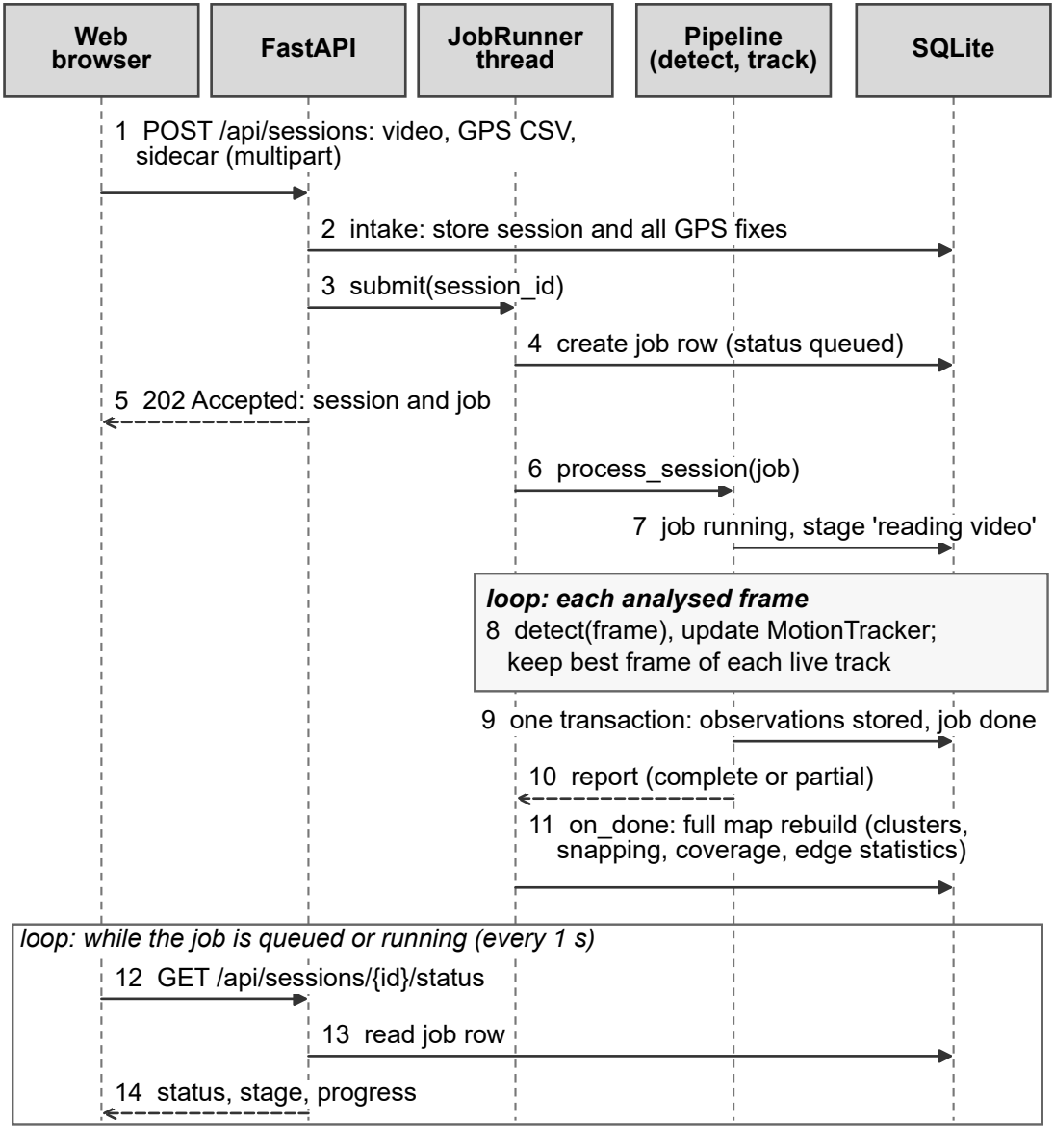

Web browser
FastAPI
JobRunner thread
Pipeline (detect, track)
SQLite
1 POST /api/sessions: video, GPS CSV, sidecar (multipart)
2 intake: store session and all GPS fixes
3 submit(session_id)
4 create job row (status queued)
5 202 Accepted: session and job
6 process_session(job)
7 job running, stage 'reading video'
loop: each analysed frame
8 detect(frame), update MotionTracker; keep best frame of each live track
9 one transaction: observations stored, job done
10 report (complete or partial)
11 on_done: full map rebuild (clusters, snapping, coverage, edge statistics)
loop: while the job is queued or running (every 1 s)
12 GET /api/sessions/{id}/status
13 read job row
14 status, stage, progress